

3 (a) State what is meant by the internal energy of a system.

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.....

..... [2]

(b) An ideal gas containing  $8.7 \times 10^{23}$  molecules is at a temperature of 300 K and a pressure of  $2.0 \times 10^5$  Pa.

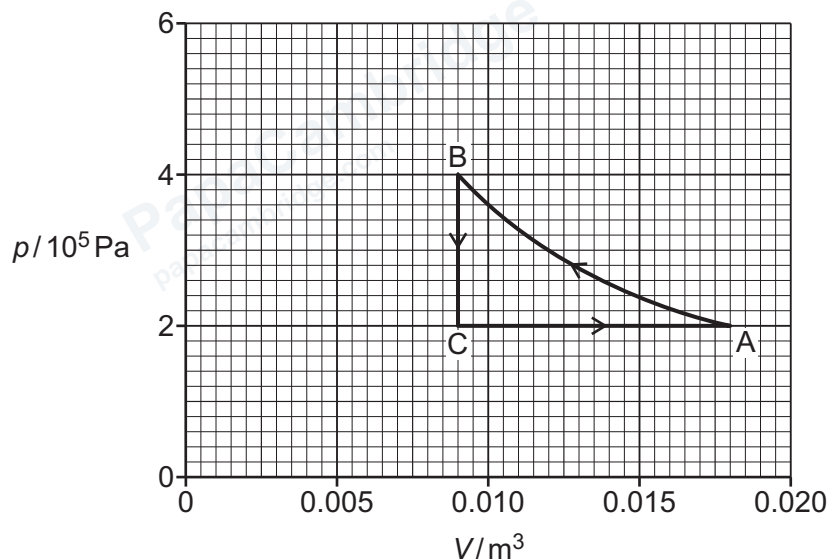
(i) Show that the volume of the gas is  $0.018 \text{ m}^3$ .

[2]

(ii) Determine the internal energy of the gas. Explain your reasoning.

internal energy = ..... J [3]

- (c) The gas in 3(b) undergoes a sequence of changes AB, BC and CA in its pressure  $p$  and volume  $V$ , as shown in Figure 3.1.



**Figure 3.1**

Determine **three** conclusions that can be drawn from Figure 3.1 about the individual and overall effects of these changes on the energy of the gas. The conclusions may be qualitative or quantitative, and may refer to any of work done, transfer of thermal energy and change in internal energy. Use the space for any working.

1 .....

2 .....

3 .....

[3]

[Total: 10]